What is Cybersecurity? Explain the CIA triad with one real-world example for each part.

Cybersecurity & The CIA Triad

- Cybersecurity: The practice of protecting networks, devices, systems, and data from unauthorized digital access, damage, or attacks.

The CIA Triad

- Confidentiality: Keeping sensitive data private so only authorized people can see it.
    - *Example*: Requiring Multi-Factor Authentication (MFA) and encryption to log into a bank account.
- Integrity: Making sure data is accurate and has not been altered or tampered with.
    - *Example*: Using a SHA-256 file hash to verify a downloaded software installer hasn't been modified by malware.
- Availability: Ensuring systems and data are up and running whenever authorized users need them.
    - *Example*: Using a DDoS protection service like Cloudflare to keep a website online during an attack.